

Vlan and Secure Switch Configuration

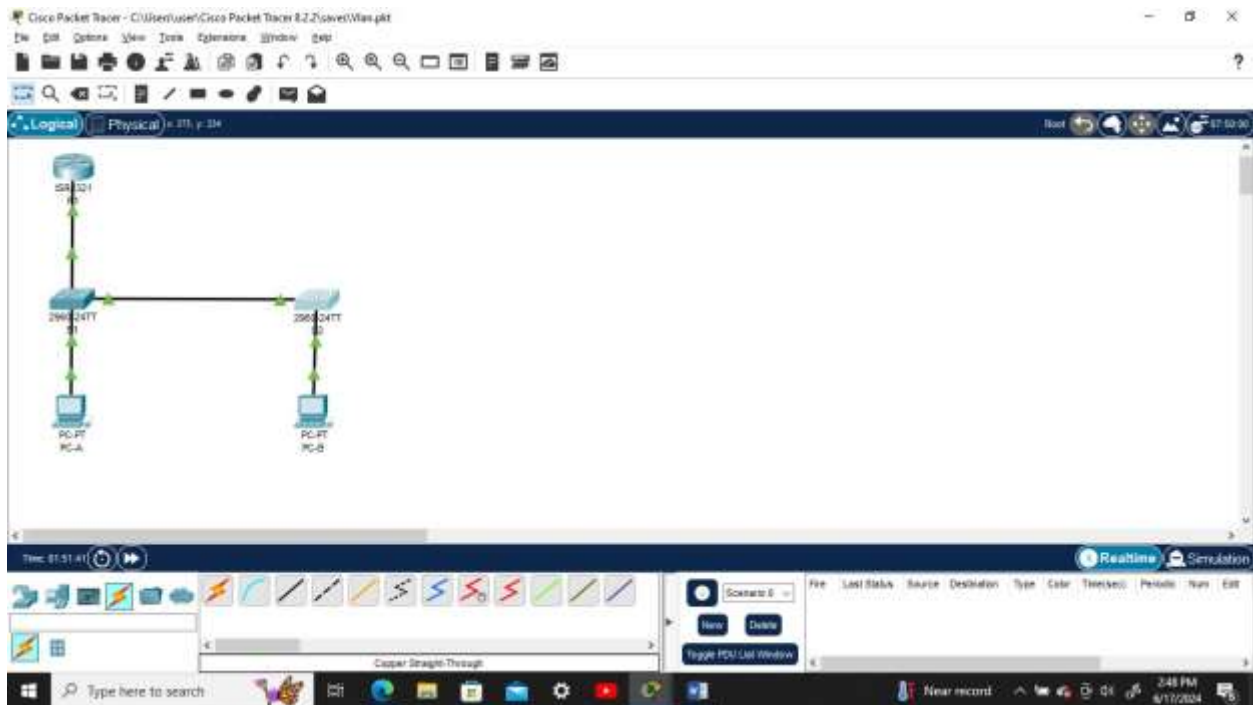
Report by Ezekiel Waweru CNS06-24049

This assignment explored how to setup a secure cloud network. Using packet tracer, I was able to realize this by setting it up and explore how it can be made possible.

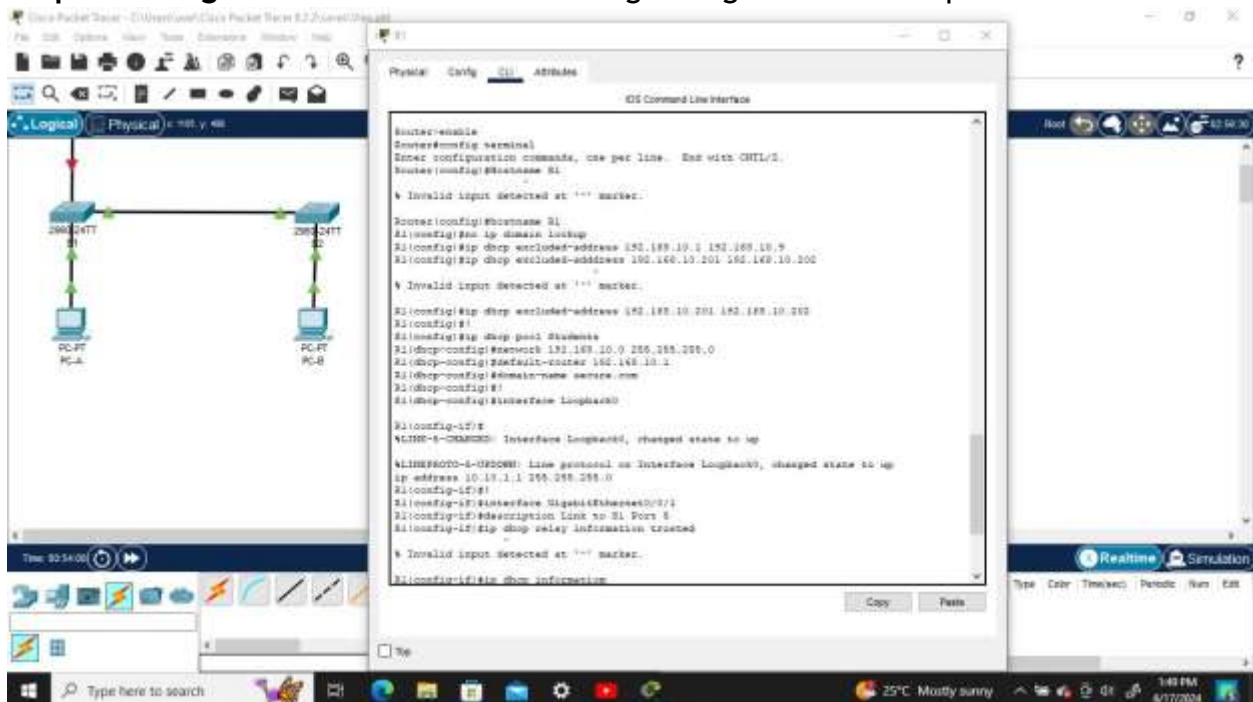
Part 1: Configure the Network Devices

Step 1: Cable the Network

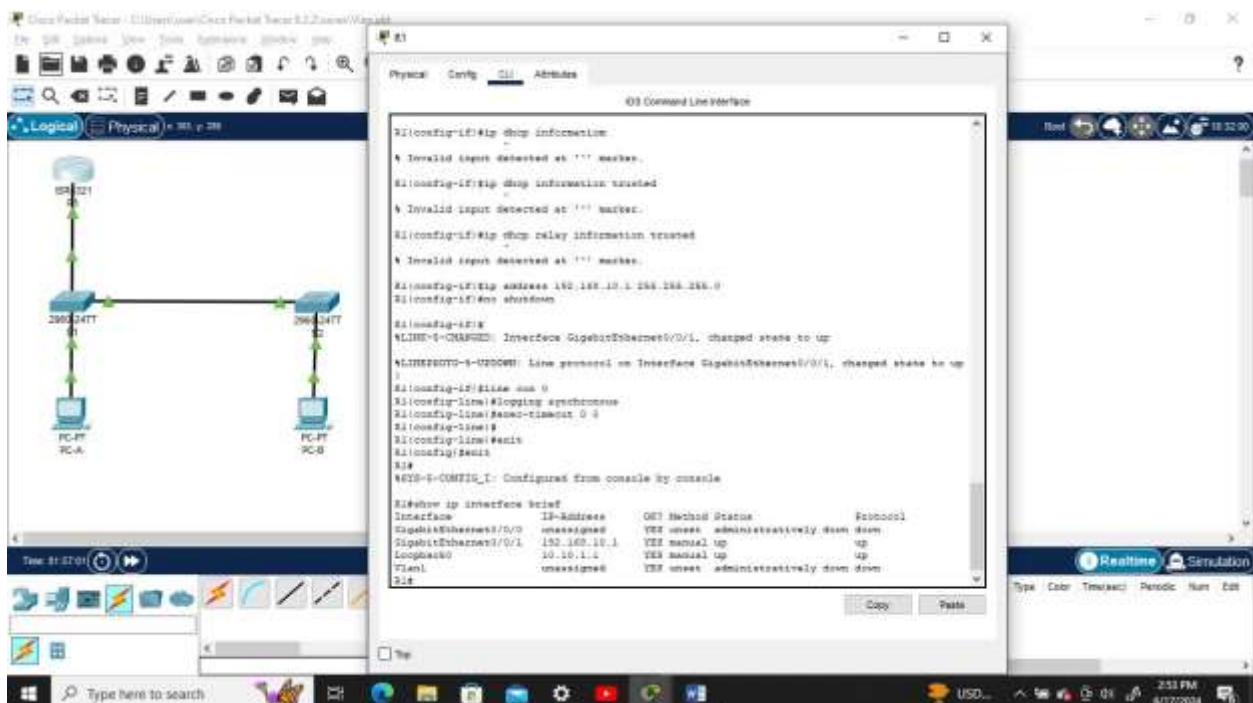
Here I cabled the network as shown in the topology and initialized the devices.



Step 2: Configure R1.I loaded the following configuration on script on R1.



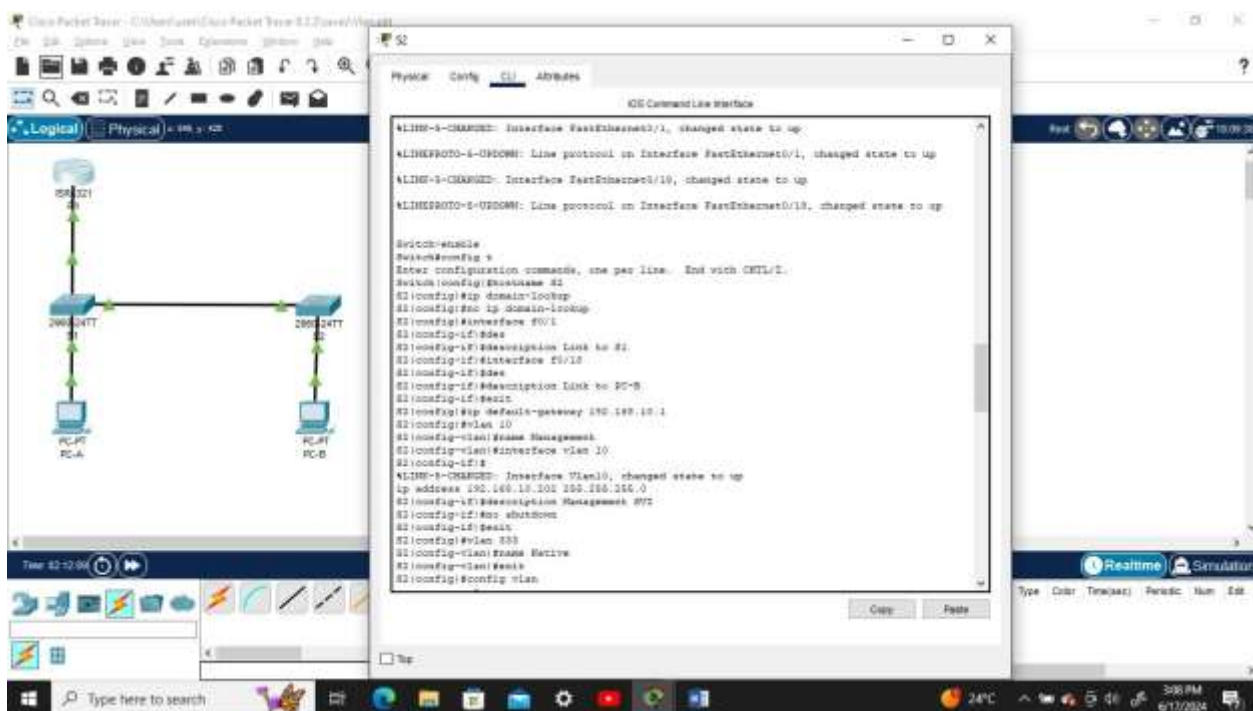
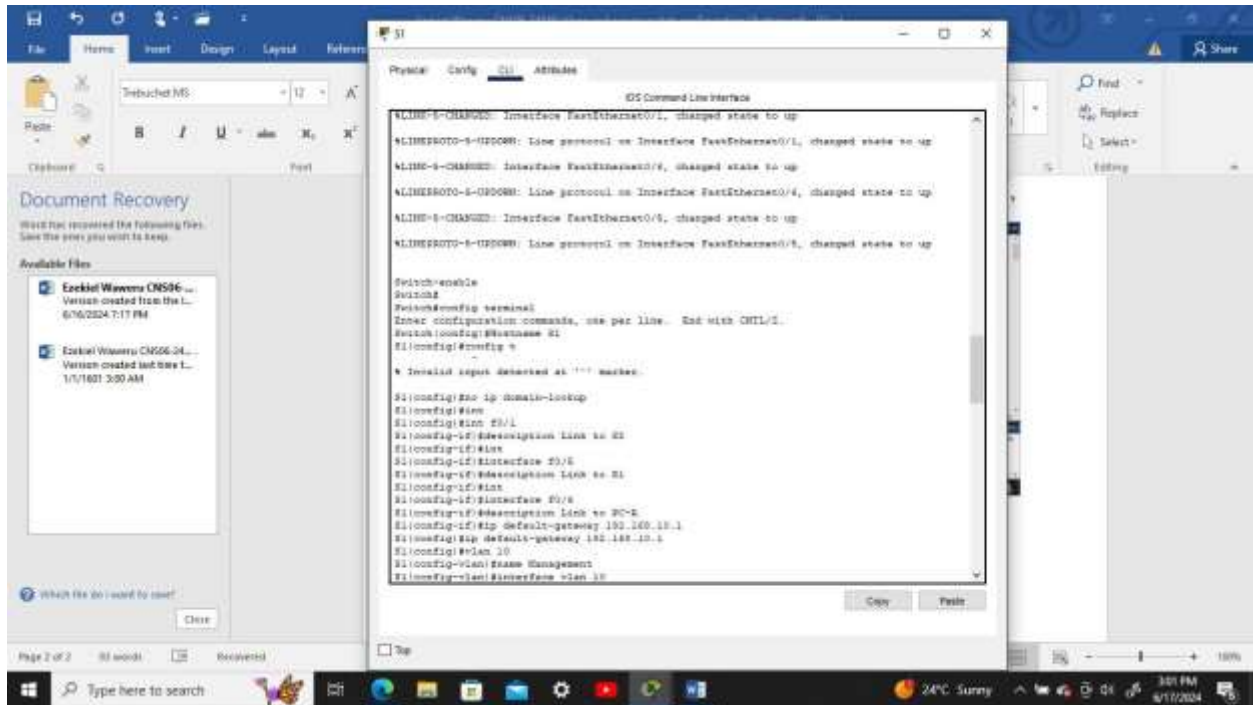
I verified the running-configuration on R1



Step 3: Configure and verify basic switch settings

I configured the hostname for switches 1 and 2. After, I prevented unwanted DNS lookups on both switches. This progressively led to configuring descriptions for the

ports that are in use in S1 and S2. Finally, I set the default-gateway for the Management VLAN to 192.168.10.1 on both switches.



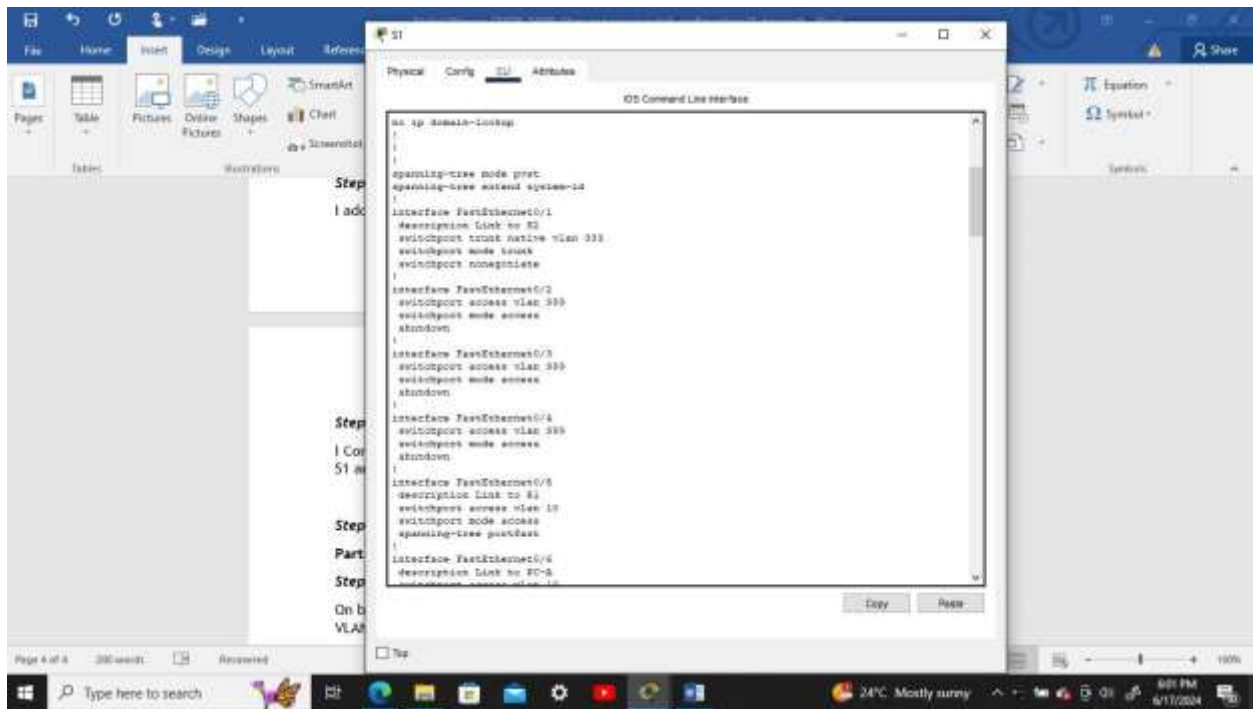
Part 2: Configure VLANs on Switches

Step 1: Configure VLAN 10

I added VLAN 10 to S1 and S2 and named the VLAN **Management**.

Step 2: Configure the SVI for VLAN 10.

I Configured the IP address according to the Addressing Table for SVI for VLAN 10 on S1 and S2. Enable the SVI interfaces and provide a description for the interface.



Step 3: Configure VLAN 333 with the name Native on S1 and S2.

Part 3: Configure Switch Security

Step 1: implement 802.1Q trunking

On both switches, I configured the trunking on F0/1 to use VLAN 333 as the native VLAN.

I then verified that the trunking is configured on both switches.

After, I disabled DTP negotiation on F0/1 on S1 and S2.

Finally, I verified with the **show interfaces** command.

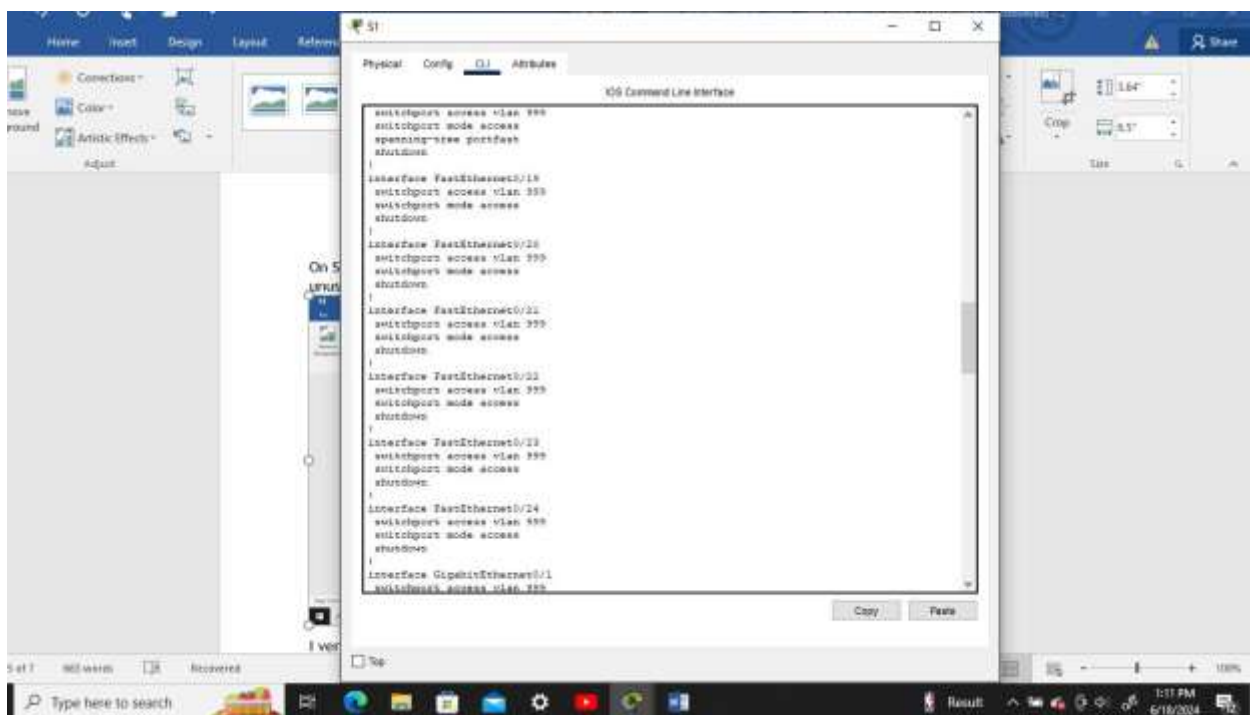
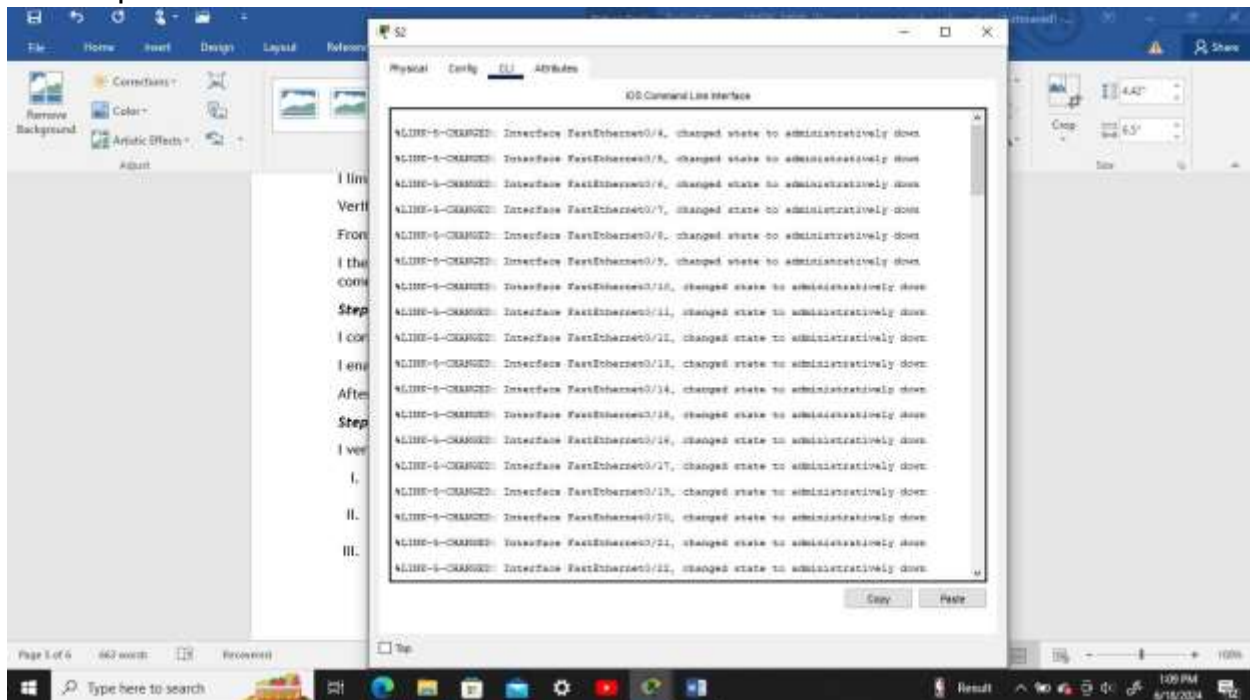
Step 2: Configure access ports.

I configured F0/5 and F0/6 as access ports that are associated with VLAN 10.

On S2, I configured F0/18 as an access port that is associated with VLAN 10/

Step3: Secure and disable unused switchports.

On S1 and S2, I moved the unused ports from VLAN 1 to VLAN 999 and disabled the unused ports.



I verified that unused ports are disabled and associated with VLAN 999 by issuing the **show** command.

Step 4: Document and implement port security features.

On S1, I issued the show port-security interface f0/6 command to display the default port security settings for interface F0/6

On S1, I enabled port security on F0/6

I verified port security on S1 F0/6

I enabled port security for F0/18 on S2, then configured the port to add MAC addresses learned on the port automatically to the running configuration.

I configured port security settings on S2 F/18

I then verified port security on S2 F0/18

Step 5: Implement DHCP snooping security

On S2, I enabled DHCP snooping and configured DHCP snooping on VLAN 10

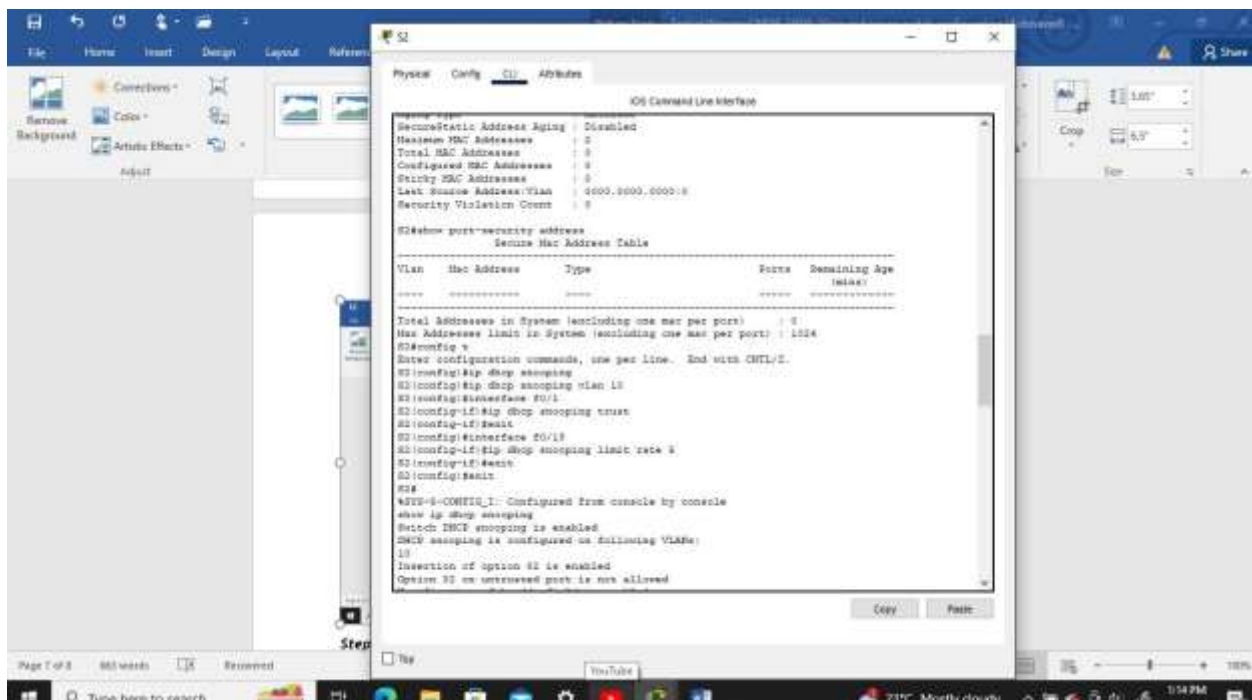
I then configured the trunk port on S2 as a trusted port

I limited the untrusted port, F18 on S2, to five DHCP packets per second

Verifying DHCP snooping was on S2 was the next step

From the command prompt on PC-B. I released and then renewed the IP address

I then verified the DHCP snooping binding using the show ip dhcp snooping binding command

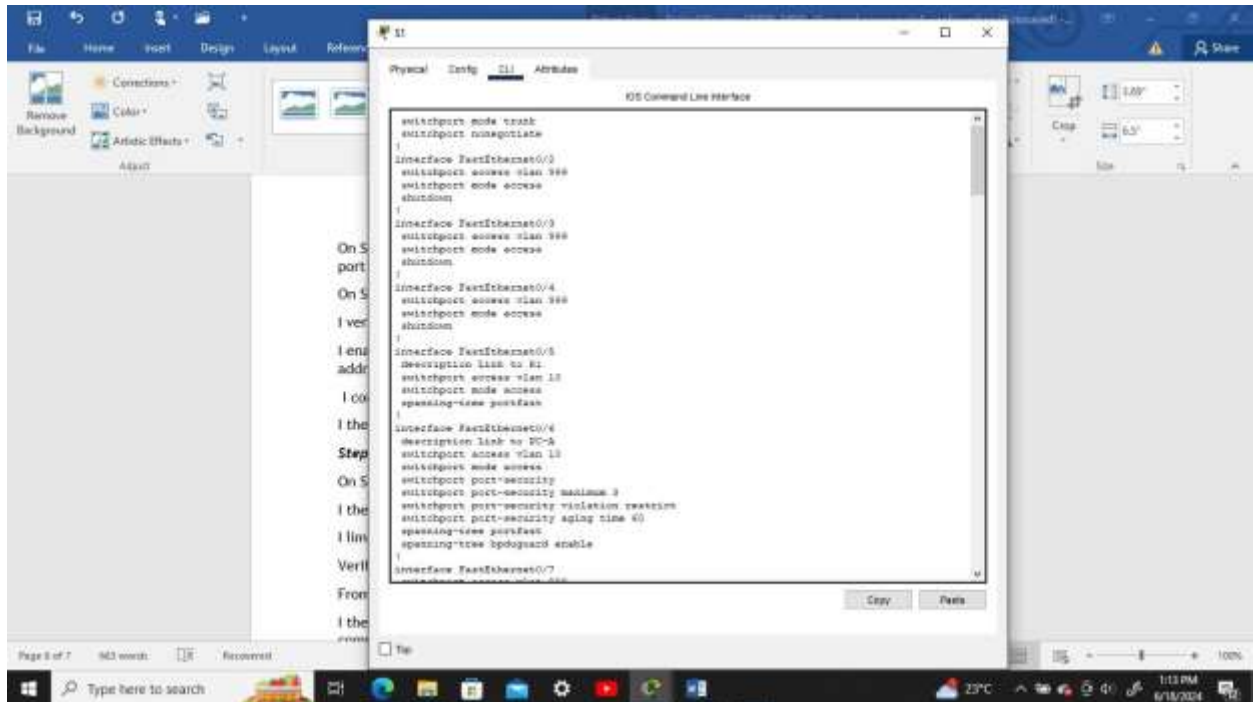


Step 6: Implement PortFast and BPDU guard

I configured PortFast on all the access ports that are in use on both switches

I enabled BPDU guard on S1 and S2 VLAN 10 access ports connected to PC-A and PC-B

After, I verified that BPDU guard and PortFast are enabled on the appropriate ports

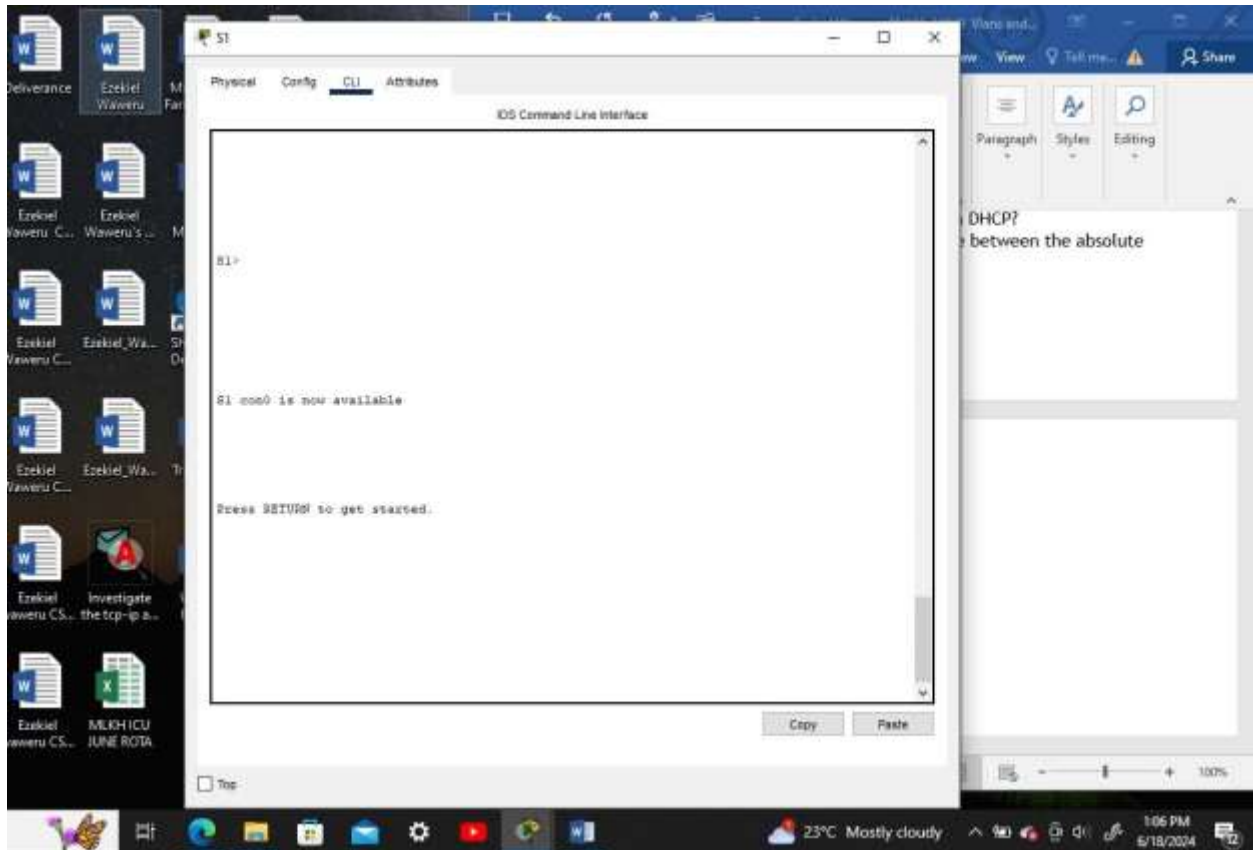


Step 7: Verify end-to-end connectivity

I verified ping connectivity between all devices in the IP Addressing Table.

- I. In reference to Port Security on S2, why is there no timer value for the remaining age in minutes when sticky learning was configured?
- II. In reference to Port Security on S2, if you load the running-config script on S2, why will PC-b on port 18 never get an IP address via DHCP?
- III. In reference to Port Security, what is the difference between the absolute aging type and inactivity aging type?

Device Configurations-Final



Conclusion

This assignment was educative as I got to learn more on how to use packet tracer and also how security in the cloud can be implemented administratively. My confidence and skill growing with every assignment, it was a fun time tackling this. Configuring the different network devices was something I enjoyed and I'm sure I will enjoy doing in the present future.